
Machine Learning Based Analog Circuit Fault Diagnosis
PDF 9 — Chapter 15: The Topology Rule
The Engineering Idea That Made This Project Different | Hybrid Intelligence

PART 1 — The Engineering Idea That Made Our Project Different

Chapter 15 — The Topology Rule Part 1 — The Engineering Idea That Made Our Project Different Before Starting In the previous chapter, we solved part of the problem. We discovered that RC_Open was being misclassified. We investigated the features. We added DC Level. Question. Problem solved? Not completely. Some confusion still remained. Question. Why? This time the answer was not Machine Learning. The answer was Circuit Topology.

This Is Where Our Project Changed

Until now, every improvement came from Machine Learning. Now, the improvement comes from Electronics. That is exactly what makes our project different from a normal classification project.

First Question

What is Topology? Many students hear this word for the first time. Actually, it is very simple. Topology means how the components are connected. Not their values. Not their names. Only their connection. Example Imagine three components. Resistor Capacitor Inductor Question. Can they be connected in different ways? Yes. Example. Series or Parallel or Series-Parallel The components are the same. Only their connections change. That is Topology.

PROFESSOR ASKS

What is Circuit Topology?

EXCELLENT ANSWER

Circuit topology refers to the way electrical components are interconnected, independent of their numerical component values.

Real Life Analogy

Imagine LEGO blocks. Question. Suppose I give you 20 LEGO pieces. Can you build different models? Yes. Car. House. Bridge. Question. Did the pieces change? No. Only their arrangement changed. Exactly what Topology means.

Now Connect It To Our Project

Question. Suppose two circuits have different topologies. Question. Can they still produce similar feature values? Yes. That was our biggest discovery. Why? Remember what Machine Learning sees. Does it see the circuit? No. Does it know where the resistor is connected? No. Does it know the schematic? No. It only sees Peak RMS Mean

Rise Time

. DC Level Just 17 numbers. Nothing more.

Very Important Statement

Machine Learning

does not understand circuits. It only understands features. Remember this sentence. It explains almost everything.

The Problem

Suppose

Circuit A

and

Circuit B

have different physical structures. But their 17 features become very similar. Question. Can SVM separate them perfectly? Not always. Because from the SVM's perspective, they look almost identical.

Real Life Analogy

Imagine two twins. Question. Can face recognition sometimes confuse them? Yes. Question. Why? Because their features are extremely similar. Exactly the same problem.

Our Realization

At this point, we understood something important. Question. Was the classifier necessarily wrong? No. Question. Were the features necessarily wrong? No. Instead, some circuit behaviours were naturally similar. That similarity came from the physics of the circuits, not from the machine learning algorithm.

PROFESSOR ASKS
Why couldn't the classifier perfectly separate every fault?
EXCELLENT ANSWER

Some fault conditions produced highly similar feature vectors because of their underlying circuit behavior. Since the classifier only receives feature values and has no knowledge of the circuit topology, certain classes naturally become difficult to distinguish. This is an excellent research-level answer. What Is The Topology Rule?

The Topology Rule

is a simple idea. It says Do not rely only on machine learning. Also use engineering knowledge about how the circuit is connected and behaves. Notice something. We did NOT replace Machine Learning. We complemented it. Very important. Why Is This Better? Suppose

Machine Learning

says Probably RC_Open Instead of blindly accepting the prediction, we ask another question. Does this prediction make sense considering the circuit topology? Now our system uses both

Machine Learning

and Engineering. That is a much stronger approach. Engineering vs Pure AI Pure AI says Trust the model. Engineering says Trust the model, but verify it using domain knowledge. That is exactly what happened in your project.

Why This Is Valuable

Many projects only apply an algorithm. Your project went one step further. It asked What does circuit theory tell us? That question made the project much stronger.

Connection to Our Project

The evolution of the project looked like this.

Initial Features

SVM

Confusion Matrix

RC_Open Errors — DC Level Added.

Remaining Ambiguity

Circuit Topology Analysis

Topology Rule

Notice every improvement came from understanding the previous weakness. Nothing was random.

Why Interviewers Like This

Suppose an interviewer asks "Why didn't you simply use a more complex model?" A weak answer is "I don't know." A strong answer is "The remaining ambiguity was caused by similarities in circuit behavior rather than limitations of the classifier. Therefore, we combined machine learning with engineering knowledge instead of simply increasing model complexity." That answer shows engineering maturity.

COMMON MISTAKES

MISTAKE: [X] Topology Rule replaces Machine Learning. Wrong. It complements it.

MISTAKE: [X] Machine Learning understands the circuit diagram. Wrong. It only sees feature vectors.

MISTAKE: [X] Every classification error can be fixed with a better algorithm. Wrong. Sometimes the limitation comes from the underlying physics.

REVISION NOTES

Remember these ideas.

Topology — Component Connections.

Machine Learning — Sees Features.

Engineering — Understands Circuits.

Topology Rule — Combine Both.

MEMORY TRICK

Remember this sentence. "Machine Learning sees numbers. Engineers see circuits. The Topology Rule uses both." One sentence. Entire chapter.

KNOWLEDGE CHECK

Level 1

- What is circuit topology?
- Why doesn't SVM understand circuit topology?
- What problem remained after adding DC Level?
- What is the main idea behind the Topology Rule?
- Does the Topology Rule replace SVM?

Level 2

- Why can two different circuit topologies produce similar feature vectors?
- How does engineering knowledge complement machine learning?
- Why wasn't using a more complex classifier the first solution?
- How did the Topology Rule improve the overall reasoning of the project?
- Why is this approach more robust than relying only on the classifier?

Professor Level

- Explain how the Topology Rule evolved from confusion matrix analysis.
- Why are some classification errors fundamentally caused by circuit physics?
- What are the limitations of feature-based classifiers in analog fault diagnosis?
- How does combining domain knowledge with machine learning improve system reliability?
- Why is the Topology Rule one of the most original contributions of your project?
- End of Chapter 15 (Part 1)
- Congratulations.
- You now understand the idea behind the Topology Rule.
- More importantly, you understand why it became necessary—not because the SVM was weak, but because machine learning alone cannot capture every aspect of circuit behavior.

Author's Note The next part will go even deeper. We'll explain: Exactly how the Topology Rule was applied in your project. Which fault classes it helps distinguish. Practical examples using your RC, RL, and RLC circuits. How to justify this contribution in a viva as an engineering enhancement rather than a post-processing hack. Why this idea makes your project more aligned with real industrial diagnostic systems.

PART 2 — Applying the Topology Rule in Our Project

Chapter 15 — The Topology Rule Part 2 — Applying the Topology Rule in Our Project Before Starting

In Part 1

we learned the idea behind the Topology Rule. Today we'll answer a much bigger question. How exactly did we apply it in our project? This is the part that most professors will enjoy because it shows you didn't simply run Python code. You actually understood the electronics.

First Question

Suppose our SVM predicts RC_Open Question. Should we blindly accept the prediction? Most beginners say Yes. Our answer was No. Why? Because

Machine Learning

can only use the information present inside the feature vector. It cannot ask questions like Does this waveform make physical sense? Only an engineer can ask that question.

Real Life Analogy

Imagine Google Maps. It suggests one route. Question. Will you always follow it? Suppose you see the road is flooded. Question. Will you still go? No. You use your own knowledge. Exactly the same idea.

Machine Learning

gives a prediction. Engineering verifies it. What Was Our Goal? Our goal was never to prove that SVM is wrong. Our goal was to make the final diagnosis more reliable. Very important. Think Like an Engineer Question. Suppose the classifier is 95% confident. Question. Should confidence alone be enough? No. Question. Why? Because a physically impossible answer is still impossible even if the classifier is confident. Example Imagine someone says This resistor has

-50 Ω

Question. Can that happen in our passive circuit? No. Physics rejects it. Exactly the same idea.

Machine Learning

must always respect engineering principles. Applying the Rule Let's understand our thinking.

Step 1

Generate the waveform. **Circuit** — Transient Response. Step 2 Extract 17 features. **Waveform** — 17 Features. Step 3 Train the SVM.

17 Features

SVM Step 4 Get the prediction. Example **Prediction** — RC_Open. Step 5 Now comes our addition. Instead of stopping, we ask another question. Does this prediction agree with the expected circuit behaviour? That is the Topology Rule.

PROFESSOR ASKS

What is the purpose of the Topology Rule?

EXCELLENT ANSWER

The Topology Rule acts as an engineering validation step. After obtaining the machine learning prediction, the predicted fault is checked against the expected physical behavior of the circuit to improve the reliability of the diagnosis. Excellent. What Does "Expected Behaviour" Mean? Question. Suppose a capacitor is open. Question. Should the waveform look exactly like a resistor short? Obviously No. Different faults produce different physical behaviour. That behaviour must be consistent with the prediction.

Engineering Knowledge

Throughout our project, we already knew how RC, RL, and RLC circuits should behave. For example, we know RC circuits typically show charging or discharging behaviour. We know RLC circuits can oscillate. We know different faults change the transient response in different ways.

The Topology Rule

simply says Use that knowledge.

Important Clarification

Many students misunderstand this idea. They think the Topology Rule changes the prediction. Not necessarily. Sometimes it simply explains why the prediction is reasonable. Sometimes it warns us that the prediction should be investigated further. Why Is This Important? Imagine a doctor. AI says Pneumonia Doctor checks the X-ray, the symptoms, the patient's history. Question. Did the doctor ignore AI? No. Question. Did the doctor blindly trust AI? Also No. The doctor combined AI with medical knowledge. Exactly what we did.

This Is Called

Hybrid Intelligence

Notice our project became a hybrid system.

Machine Learning

+

Engineering Knowledge

Final Diagnosis

This is much stronger than Machine Learning alone.

Why Industries Like This

In real industries, critical decisions are rarely made using only AI. Usually, AI provides a recommendation. An engineer verifies that recommendation. Your project follows that philosophy. That is one reason it is more realistic.

Connection to Our Project

Our final pipeline became **LTspice** — CSV Dataset.

Feature Extraction

17 Features

SVM — Prediction.

Topology Rule

Final Diagnosis

Notice

Machine Learning

is not removed. It is strengthened. Does Every Prediction Need It? Question. Suppose the classifier is predicting a fault that has very distinctive features. Question. Do we always need extra checking? Not necessarily.

The Topology Rule

is most valuable for faults that naturally produce similar feature vectors and are therefore more difficult to distinguish.

Interview Question

PROFESSOR ASKS

Why didn't you simply use a Deep Neural Network instead of introducing the Topology Rule?

EXCELLENT ANSWER

A more complex model cannot create information that is absent from the input features. The remaining ambiguity was caused by similarities in circuit behavior rather than insufficient model complexity. Therefore, incorporating engineering knowledge through the Topology Rule was a more meaningful solution. This answer is excellent. It shows you understand both

Machine Learning

and Electronics.

Biggest Lesson

This chapter teaches one important lesson.

Machine Learning

is not a replacement for engineering. It is a tool that becomes much more powerful when combined with engineering knowledge. That lesson applies far beyond this project.

COMMON MISTAKES

MISTAKE: [X] Topology Rule means ignoring Machine Learning. Wrong.

MISTAKE: [X] Machine Learning must always be trusted. Wrong.

MISTAKE: [X] A larger neural network would automatically solve every ambiguity. Wrong.

MISTAKE: [X] Engineering knowledge is unnecessary once AI is trained. Wrong.

REVISION NOTES

Remember

the complete flow.

Machine Learning — Prediction.

Engineering — Validation.

Together — Reliable Diagnosis.

MEMORY TRICK

Remember this sentence. "AI predicts. Engineering verifies. Together they diagnose." One sentence. Entire philosophy.

KNOWLEDGE CHECK

Level 1

- What is the purpose of the Topology Rule?
- Does it replace the SVM?
- Why is engineering validation useful?
- What is hybrid intelligence?
- Where is the Topology Rule applied in the pipeline?

Level 2

- Why can't machine learning alone solve every ambiguity?
- Why is domain knowledge important in analog fault diagnosis?
- How does the Topology Rule improve reliability?
- Why isn't increasing model complexity always the best solution?
- How does this approach resemble industrial diagnostic systems?

Professor Level

- Explain how the Topology Rule complements SVM in your project.
- Why is feature similarity sometimes caused by circuit physics rather than classifier limitations?
- Discuss the role of domain knowledge in machine learning-based fault diagnosis.
- Why is your approach an example of hybrid intelligence?
- If you were extending this project to industrial-scale fault diagnosis, how would the Topology Rule evolve?
- End of Chapter 15
- Congratulations.
- You have now completed one of the most original sections of the Project Bible.
- This chapter demonstrates that your project is not just about training an SVM—it is about combining machine learning with engineering reasoning to build a more reliable fault diagnosis system.

Author's Note (Important) While preparing this Bible, I noticed something significant. We've now completely documented: [OK] Electronics theory [OK] LTspice workflow [OK] Dataset generation [OK] Feature extraction [OK] All 17 features [OK] SVM theory and implementation [OK] GridSearchCV [OK] Model evaluation [OK] RC_Open investigation [OK] DC Level [OK] Topology Rule From this point onward, the remaining chapters will mostly focus on: Your complete Python code, explained line by line. Dashboard implementation and every widget/button. ANN implementation and comparison with SVM. Future scope, limitations, viva questions, interview questions, and presentation strategy. The conceptual foundation of the project is now complete. The remaining sections are about understanding the implementation and presenting it effectively.